Koa Shen

Mechanical Engineering | Embedded Controls, Mechatronics & Motion Systems | Summer 2027

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EDUCATION

University of California, Santa Barbara Expected June 2028

B.S. Mechanical Engineering | Major GPA: 3.97 | Cumulative GPA: 3.94

- Dean's Honors every quarter. Relevant coursework: Circuits, Dynamics, MATLAB; Dynamical Systems and Machine Learning in progress.

ENGINEERING EXPERIENCE

Mechanical Engineering Intern Summer 2026 - Present

SLAC National Accelerator Laboratory, LCLS | Menlo Park, CA; extended part-time remote

- Built a Python/Drake digital twin for a 30-DOF, 305-body motion assembly; millisecond collision checks identified at least five physical pinch points and informed mechanical redesigns.
- Created 15 reusable motion-stage models using CAD import, collision geometry, and visualization; framework adoption is underway across three LCLS assemblies.
- Used kinematic simulation and load checks to guide optics-stage packaging, mounts, and alignment redesign around sensitive $100k+ detectors.
- Produced Solid Edge designs and PDM-controlled documentation with GD&T; path planning and EPICS controls integration remain in development.

Fabrication Team Lead Fall 2024 - Present

Gaucho Racing, Formula SAE Electric | UC Santa Barbara

- Led 15 teammates through driver safety-system design, manufacturing, and installation; GR26 passed FSAE technical inspection fully rules-compliant.
- CNC-machined 33 suspension parts from 4130 steel and 7075-T6 aluminum, holding critical bearing fits within 0.0005 in.
- Own chassis and ergonomics packaging, coordinating welded structures, assembly interfaces, and manufacturing across vehicle subteams.

Maintenance Technician Summer 2025

Exploratorium | San Francisco, CA

- Serviced 100+ exhibits; reverse-engineered and redesigned encoder mounts and mechanisms for reliability and repeated use, documenting CAD, CAM, and remanufacturing details.

CONTROLS & ROBOTICS PROJECTS

Closed-Loop Cycloidal Actuator Summer 2026 - Present

- Designed a compact 15:1 reducer for a NEMA 17 stepper and built PlatformIO C++ firmware for step/dir control, real-time telemetry, and a TMC2209 driver.
- Validated 12-bit AS5600 motor-shaft encoder feedback on a single-joint bench setup; firmware compensates position error and detects missed steps. Multi-joint sensing and control remain planned.

Formula SAE Steering-Wheel Controls Integration 2024 - 2026

- Managed GR25 wheel design and supported GR26 integration; combined buttons, potentiometers, display, and quick-release hardware in a driver interface. Iterative design and FEA reduced assembly mass from 5 lb to 1.7 lb.

TECHNICAL SKILLS

Embedded Controls: C/C++, PlatformIO, Arduino, step/dir, TMC2209, AS5600 encoder feedback, closed-loop position correction

Modeling & Analysis: Python, Drake, MATLAB, kinematics, dynamics, collision detection, FEA

Hardware Integration: Sensors, stepper motors, electromechanical packaging, CAD, GD&T, prototyping, test